
Stage 1: Synthetic dataset creation
Synthea
Stage 2: Raw datasets
Patients
Encounters
Organizations
Providers
Devices
Conditions
Allergies
Observations
Procedures
Imaging Studies
Immunizations
Medications
Payers
Payer Transitions
Careplans
Stage 3: Data cleaning
Format cleaning
JSON to CSV: Patients, Medications, Imaging Studies, Allergies,
Payers, Payer Transitions
Outliers
Missing values
Patients 4652 · Encounters 79138 · Conditions 43098 · Medications 24129
Procedures 31088 · Allergies 533 · Organizations 184 · Payers 5
Payer Transitions 236 · Careplans 2186 · Devices 78
Duplicates
Observations: 442
Validation
Range validation: removed 5 invalid records from Medications
Relationship validation
Stage 4: Data profiling
Numerical statistics
Column profiles
Data dictionary
Stage 5: Data staging
PostgreSQL database
Star schema